-: 12th BOARD HACKER JUGAAD NOTES :-

RELATIONS :-

A relation R from a non-empty set A to a non empty set B is a subset of the Cartesian product A × B. The set of all first elements of the ordered pairs in a relation R from a set A to a set B is called the domain of the relation R. The set of all second elements in a relation R from a set A to a set B is called the range of the relation R. The whole set B is called the codomain of the relation R. Note that range is always a subset of Codomain.

Ex :-

$A = \{2, 3\}$

$B = \{1, 4, 5\}$

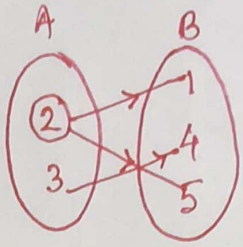
$A \times B = \{(2,1), (2,4), (2,5), (3,1), (3,4), (3,5)\}$

↓ Cartesian product

Subset

Ex :- $R = \{(2,1), (2,5), (3,4)\}$

$R : A \to B$



TYPES OF RELATIONS :-

Total number of Relations defined from set A to set B $= 2^{n(A) \cdot n(B)}$

A relation R in a set A is called

i) reflexive, if $(a,a) \in R$, for all $a \in A$

ii) Symmetric, if $(a,b) \in R \Rightarrow (b,a) \in R$, for all $a, b \in A$

iii) Transitive, if $(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$, for all $a, b, c \in A$.

Ex :-

$n(A) = 3$

$n(B) = 2$

no. of Relations from A to B $= 2^6$

No. of Relations from B to A $= 2^6$.

Types of Relations :-

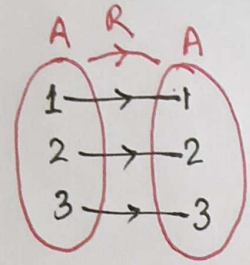
On set A / in set A

$R : A \to A$

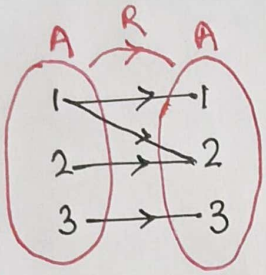
Maths :- If $'a' \in A$ then $(a,a) \in R$

① Reflexive :-

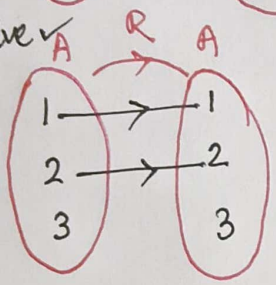
Some elem. khud se relate hoi



Reflexive ✓



Reflexive ✓

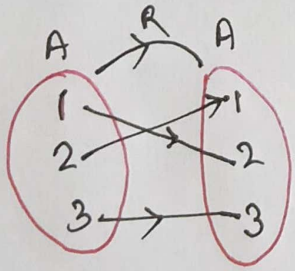


Reflexive ✗

② Symmetric :-

If $(a,b) \in R \Rightarrow (b,a) \in R$

Ex:-



Symm. ✓

OR,

$R = \{(1,2)\ (2,1)\ (3,3)\}$

③ Transitive :-

If $(a,b) \in R$ & $(b,c) \in R$
$\Rightarrow (a,c) \in R$

Ex:-

$R = \{(1,1)\ (1,2)\ (2,1)\ (2,3)\}$

$(1,1)(1,2) \to (1,2)$. not Trans

$(1,2)(2,1) \to (1,1)$

$(1,2)(2,3) \to (1,3)$

$(2,1)$ $(1,1)$

$(2,1)$ $(1,2)$

IMPORTANT NOTES :-

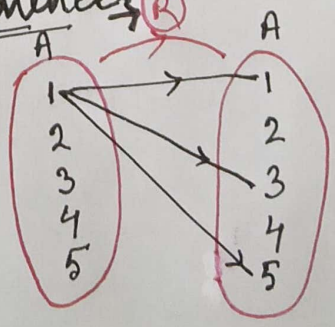
A relation R in a set A is called an **equivalence** relation, if it is reflexive, Symmetric and transitive.

The set of all elements related to any element $a \in A$ is denoted by $[a]$

i.e., $[a] = \{x \in A : (x,a) \in R\}$

Here, $[a]$ is called an equivalence class of $a \in A$.

Equivalence ;



Equivalence

Ex:- $A = \{1, 2, 3, 4, 5\}$

$R = \{(a, b) : a + b = \text{even}, \ a, b \in A\}$

Equivalence class of elem. of '1' $= [1] = \{1, 3, 5\}$

TYPES OF FUNCTION

A relation / from a set A to a set B is called a function if

i) for each $a \in A$, there exists same $b \in B$ such that $(a, b) \in f$ i.e., $f(a) = b$

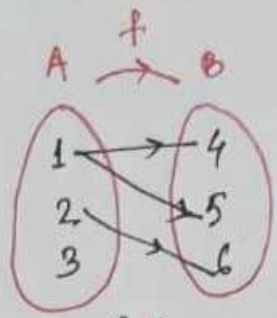
ii) $(a, b) \in f$ and $(a, c) \in f \Rightarrow b = c$

A function $f: A \to B$ is called

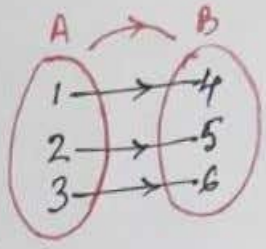
i) One - One or injective function, if distinct elements of A have distinct images i.e., for $a, b \in A$. $f(a) = f(b) \Rightarrow a = b$

ii) Onto or Surjective function, if for every elements $b \in B$, there exists some $a \in A$ such that $f(a) = b$.

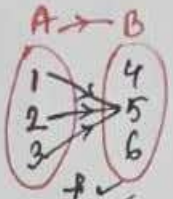
A function $f: A \to B$ is called bijective function, if it is both one - one and Onto function.



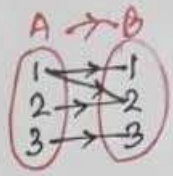
R ✓
f ✗



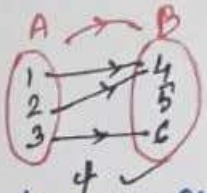
f ✓ " One - One "
R ✓ • injective mapping



f ✓ (many - one)
or (not one-one)



f ✗

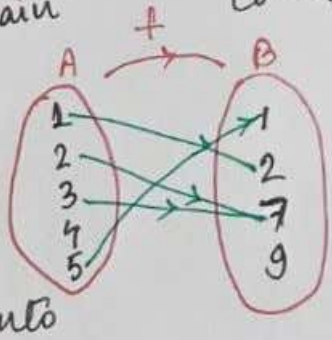


f (many - one)

Ex:-

$f : (A) \longrightarrow (B)$

domain ↙ ↘ Co-domain



⇐

- not onto
- into
- non - surjective

domain of 'f' $= A = \{1, 2, 3, 4, 5\}$

Co-domain of 'f' $= B = \{1, 2, 7, 9\}$

range of 'f' $= \{1, 2, 7\}$

Note:-

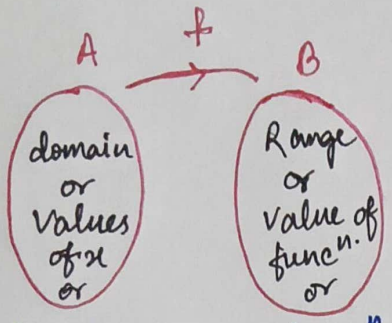
Range = Co-domain

⇓

- Onto func^n.
- Surjective

[Right side wale set ko sabhi elem. related ∈ ,]

Range ⊆ Co-domain



Pre-image func^n. image or image

Note:-

One-One + Onto

↓ ↓

inj + Surj

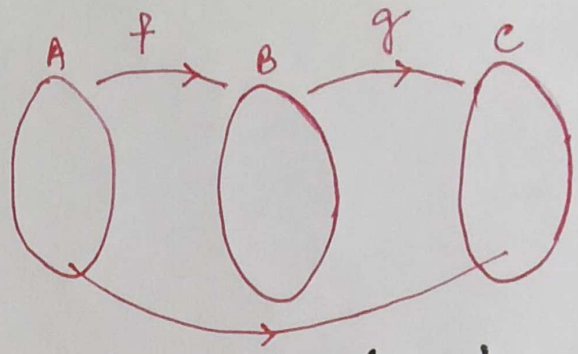
↓↓

• Bijective

COMPOSITE OF FUNCTIONS

Let $f : A \to B$ and $g : B \to C$ be two functions. Then, the composition of f and g, denoted by gof, is defined as the function $gof : A \to C$ given by

$$gof(x) = g(f(x)), \forall x \in A$$



$$gof(x) = g(f(x))$$

Ex:-

$f : R \to R$, $f(x) = x^2 + 1$

$g : R \to R$, $g(x) = x^3 + x + \frac{1}{2}$

Then,

$$g(f(x)) = (f(x))^3 + f(x) + \frac{1}{2}$$

$$= \left((x^2+1)^3 + (x^2+1) + \frac{1}{2}\right)$$

Ans.